### 2.4.1 Splitting Methods

Let $H(\boldsymbol{q},\boldsymbol{p}) = H_1(\boldsymbol{q},\boldsymbol{p}) + H_2(\boldsymbol{q},\boldsymbol{p})$ have flow map $\mathscr{F}_h$, and indicate by $\mathscr{F}_h^1$, $\mathscr{F}_h^2$ the flow maps for the systems with Hamiltonians $H_1$, $H_2$, respectively. The proposal is that the map

$$\mathscr{G}_h = \mathscr{F}_h^1 \circ \mathscr{F}_h^2$$

is an approximation of $\mathscr{F}_h$. For this to be a first order method, we need at least

$$\|\mathscr{G}_h(\boldsymbol{u}) - \mathscr{F}_h(\boldsymbol{u})\| \leq C(\boldsymbol{u})h^2.$$

Expand $\mathscr{F}_h(\boldsymbol{u})$ in a Taylor series:

$$\mathscr{F}_h(\boldsymbol{u}) = \boldsymbol{u} + h\boldsymbol{J}\nabla H + \mathcal{O}(h^2) = \boldsymbol{u} + h(\boldsymbol{J}\nabla H_1 + \boldsymbol{J}\nabla H_2) + \mathcal{O}(h^2). \tag{2.24}$$

On the other hand,

$$\mathscr{F}_h^1(\boldsymbol{u}) = \boldsymbol{u} + h\boldsymbol{J}\nabla H_1(\boldsymbol{u}) + \mathcal{O}(h^2), \qquad \mathscr{F}_h^2(\boldsymbol{u}) = \boldsymbol{u} + h\boldsymbol{J}\nabla H_2(\boldsymbol{u}) + \mathcal{O}(h^2).$$

Composing the maps we have

$$\mathscr{F}_h^1 \circ \mathscr{F}_h^2(\boldsymbol{u}) = \boldsymbol{u} + h\boldsymbol{J}\nabla H_2(\boldsymbol{u}) + h\boldsymbol{J}\nabla H_1(\boldsymbol{u} + h\boldsymbol{J}\nabla H_2(\boldsymbol{u})) + \mathcal{O}(h^2).$$

Assuming $H_1$ is $C^2$, we have

$$\mathscr{F}_h^1 \circ \mathscr{F}_h^2(\boldsymbol{u}) = \boldsymbol{u} + h(\boldsymbol{J}\nabla H_2(\boldsymbol{u}) + \boldsymbol{J}\nabla H_1(\boldsymbol{u})) + \mathcal{O}(h^2),$$

which agrees with the expansion (2.24) of the flow map through the terms of first order. Thus the splitting method does indeed provide a second order local approximation of the flow map.

*Example 2.4* Let

$$H_1(\boldsymbol{q},\boldsymbol{p}) = \boldsymbol{p}^T\boldsymbol{M}^{-1}\boldsymbol{p}/2, \qquad H_2(\boldsymbol{q},\boldsymbol{p}) = U(\boldsymbol{q}),$$

then a splitting method for $H = H_1 + H_2$ may be obtained by determining the flow maps for each of the two parts. For $H_1$ we have the differential equations

$$\dot{\boldsymbol{q}} = \boldsymbol{M}^{-1}\boldsymbol{p}, \qquad \dot{\boldsymbol{p}} = \boldsymbol{0}.$$

The second equation tells us that $p$ is constant, so it is fixed at its initial value, whereas the first equation says that $\boldsymbol{q}$ evolves on a linear path, hence its flow map is

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{p}, \\ \boldsymbol{P} &= \boldsymbol{p}. \end{aligned}$$

Similarly, the flow map of the system with Hamiltonian $H_2 = U$ is

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}). \end{aligned}$$

The composition of these maps simplifies to

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\boldsymbol{P}, \\ \boldsymbol{P} &= \boldsymbol{p} - h\nabla U(\boldsymbol{q}), \end{aligned}$$

which is precisely the Symplectic Euler method. Similarly, composing the same two maps in the opposite order gives the adjoint Symplectic Euler method.

*Example 2.5 (Verlet is Symplectic)* For the Symplectic Euler method $\mathscr{G}_h$ and its adjoint method $\mathscr{G}_h^\dagger$, consider the composition

$$\mathcal{K}_h := \mathscr{G}_{h/2}^\dagger \circ \mathscr{G}_{h/2}.$$

Let

$$\begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix} = \mathscr{G}_{h/2} \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{p} \end{bmatrix},$$

i.e.,

$$\bar{\boldsymbol{q}} = \boldsymbol{q} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \bar{\boldsymbol{p}} = \boldsymbol{p} - (h/2)\nabla U(\boldsymbol{q}).$$

and set

$$\begin{bmatrix} \boldsymbol{Q} \\ \boldsymbol{P} \end{bmatrix} = \mathscr{G}_{h/2}^\dagger \begin{bmatrix} \bar{\boldsymbol{q}} \\ \bar{\boldsymbol{p}} \end{bmatrix},$$

i.e.,

$$\boldsymbol{Q} = \bar{\boldsymbol{q}} + (h/2)\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \qquad \boldsymbol{P} = \bar{\boldsymbol{p}} - (h/2)\nabla U(\boldsymbol{Q}).$$

The composition simplifies to

$$\begin{aligned} \bar{\boldsymbol{p}} &= \boldsymbol{p} - \frac{h}{2}\nabla U(\boldsymbol{q}), \\ \boldsymbol{Q} &= \boldsymbol{q} + h\boldsymbol{M}^{-1}\bar{\boldsymbol{p}}, \\ \boldsymbol{P} &= \bar{\boldsymbol{p}} - \frac{h}{2}\nabla U(\boldsymbol{Q}). \end{aligned}$$